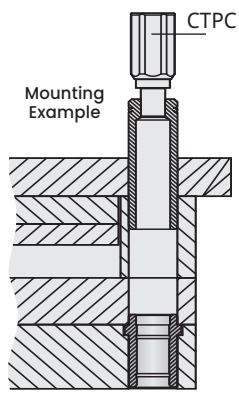


Centring Bush

Code: **G28**

d1	L	d2	M	L1
20	30	16	M8	2.5
	40			
	60			
	80			
	100			
	120			
	140			
	160			
26	30	21	M10	2.5
	40			
	60			
	80			
	100			
	120			
	140			
	160			
	180			
30	40	25	M12	2.5
	60			
	80			
	100			
	120			
	140			
	160			
	180			
	200			
	240			
42	40	33	M12	4.5
	60			
	80			
	100			
	120			
	140			
	160			
	180			
	200			
	220			
	260			
	300			
54	60	43	M12	4.5
	80			
	120			
	160			
	200			
	240			
	280			

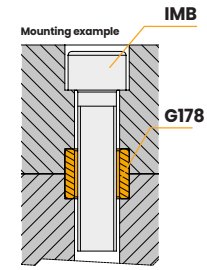


Order: **G28.** d1 x L

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

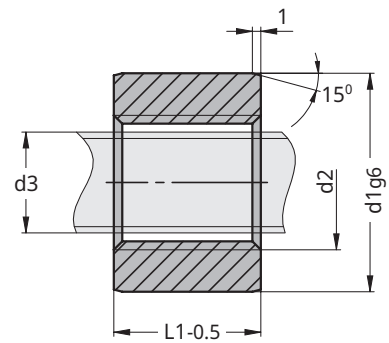
NEW





Centering Sleeve

Code: **G178**



Order Code	d1	d2	d3	L1
G178.12	12	M8	M6	10
G178.14	14	M10	M8	11.5
G178.16	16	M12	M10	13.5
G178.20	20	M16	M12	15.5
G178.26	26	M16	M14	17.5
G178.30	30	M20	M16	19.5
G178.36	36	M24	M20	19.5
G178.42	42	M30	M24	19.5



TECINMAR **G178** Centering Sleeve facilitates the precise alignment of various plates in relation to each other. Positioned above the screw connection, it helps to save space, offering increased flexibility in mould design. Additionally, the integrated pull-out thread makes disassembly much easier and more efficient.

Features:

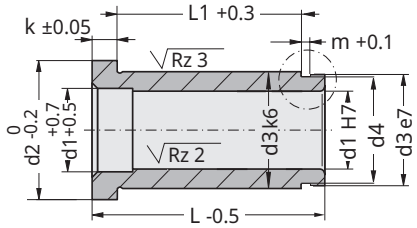
* Pull-out thread for effortless disassembly
* Compact assembly within the build-up screw connection, saving space
* 15º lead-in chamfer for easier and faster mounting
* Highly versatile for a wide range of applications
* Maximum service temperature of 200ºC

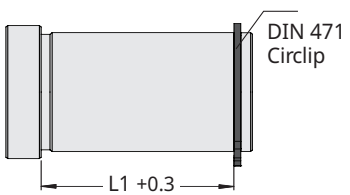


Guide Bush - Headed

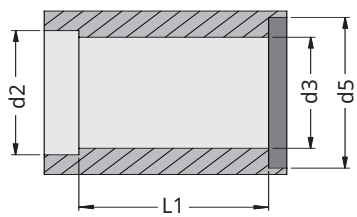
Code: **G49**

(with circlip slotted)





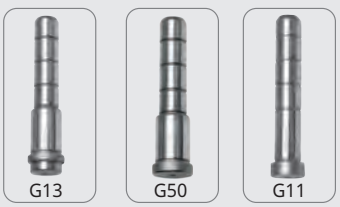
Mounting Dimensions



Order: **G49.** d1 x L

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

Compatible Pillars



For general usage in injection mould guide systems - it prevents the bush protrude in free rotating plates (**H6** centre fixing).

d1	L	d3	d2	k	L1	m	d4	d5	Circlip
12	27	18	25	6	18	1.3	17	27	18
	36				27				
	46				37				
	56				47				
14 15	27	20	25	6	18	1.3	19	30	20
	36				27				
	46				37				
	56				47				
	66				57				
	76				67				
	86				77				
	96				87				
	116				107				
16	27	22	27	6	18	1.3	21	32	22
	36				27				
	46				37				
	56				47				
18 20	27	26	31	6	18	1.3	25	36	26
	36				27				
	46				37				
	56				47				
	66				57				
	76				67				
	86				77				
	96				87				
	116				107				
	136				127				
22 24	27	30	35	6	17	1.6	29	41	30
	36				26				
	46				36				
	56				46				
	66				56				
	76				66				
	86				76				
	96				86				
	116				106				
	136				126				
	156				146				

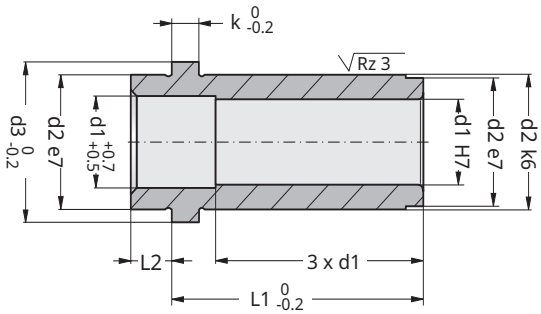
d1	L	d3	d2	k	L1	m	d4	d5	Circlip
30 32	27	42	47	6	15	1.8	40	44	42
	36				24				
	46				34				
	56				44				
	66				54				
	76				64				
	86				74				
	96				84				
	116				104				
	136				124				
	156				144				
	176				164				
	196				184				
32 40	36	54	60	10	19	2.1	51	67	54
	46				29				
	56				39				
	66				49				
	76				59				
	86				69				
	96				79				
	116				99				
	136				119				
	156				139				
	176				159				
	196				179				
	246				229				
50	76	66	72	10	58	2.6	62	79	65
	96				78				
	116				98				
	136				118				
	156				138				
	196				178				
60	96	80	86	20	68	2.6	77	95	80
	116				88				
	136				108				
	156				128				
	196				168				
	246				218				

The **G49** is delivered with the circlip included - no need to order the circlip separately.



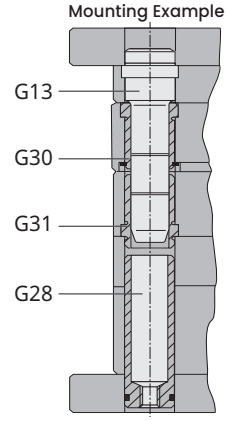
Code: **G31**

Guide Bush with Centring Collar



d1	L1	d2	d3	k	L2
14 15	17 22 27 36 46 56 66 76 86 96 116	20	25	6	9
18 20	17 22 27 36 46 56 66 76 86 96 106 116 136	26	31	6	9
22 24	17 22 27 36 46 56 66 76 86 96 106 116 136 156	30	35	6	9
30 32	27 36 46 56 66 76 86 96 106 116 126 136 156 196	42	47	6	9
40 42	46 56 66 76 86 96 116 136 156 196 246	54	60	10	12

Mounting Example



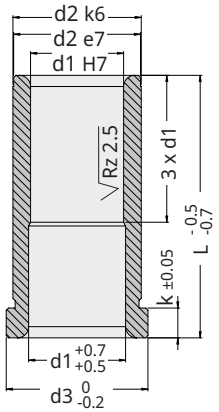
3D CAD - Formats: CATIA, STEP, IGES 3D, X_T

Order: **G31.** d1 x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

for Turkish Mould Standard



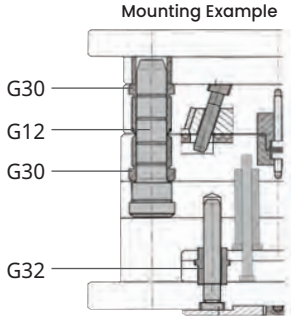


Guide Bush - Headed

Code: **G30**

d1	L	d2	d3	k
10	17	14	16	3
	27			
	36			
	46			
	56			
12	26	16	19	4
	36			
	46			
	56			
	66			
14	26	20	25	6
	36			
	46			
	56			
	66			
	76			
16	26	24	28	6
	36			
	46			
	56			
	66			
	76			
	86			
18	26	26	31	6
	36			
	46			
	56			
	66			
	76			
	86			
20	26	28	32	6
	36			
	46			
	56			
	66			
	76			
	86			
	96			
25	26	34	38	8
	36			
	46			
	56			
	66			
	76			
	86			
	96			
	106			
30	36	39	43	10
	46			
	56			
	66			
	76			
	86			
	96			
	106			
	116			
40	46	54	58	10
	56			
	66			
	76			
	86			
	96			
	106			
	116			
	126			
	136			
	146			

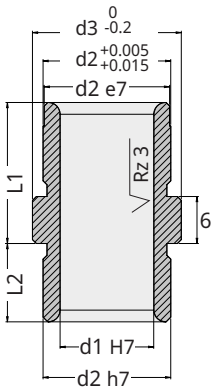
Mounting Example



Order: **G30.** d1 x L

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC



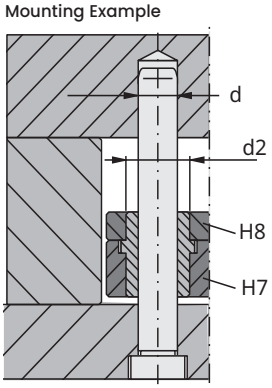


Code: **G32**

Guide Bush with Centre Collar

d1	L1	d2	d3	L2
12	16	16	19	12
14	16	20	25	12
16	16	24	28	12
18	16	26	31	12
20	16	28	32	12
24	21	34	38	16
25	21	34	38	16
30	21	39	43	16

Mounting Example



G32 Bushes ensuring the movements of ejector plates in injection moulds be rigid and precise.

Order: **G32.** d1 x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

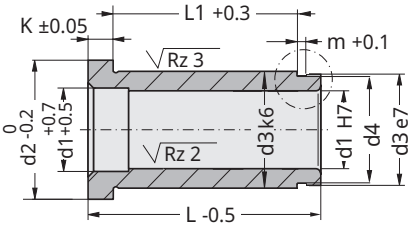
for Turkish Mould Standard

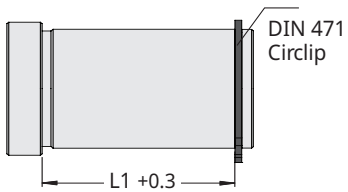


Guide Bush - Headed

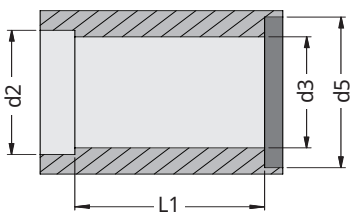
Code: **G77**

(with circlip slotted)





Mounting Dimensions



Order: **G77.** d1 x L

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

Compatible Pillars



For general usage in injection mould guide systems - it prevents the bush protrude in free rotating plates (**H6** centre fixing).

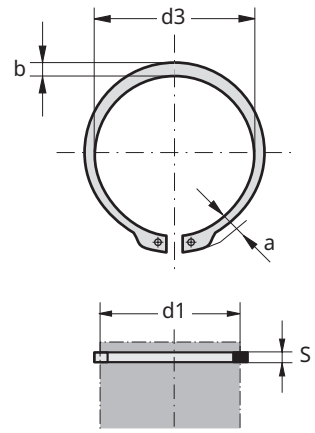
d1	L	d3	d2	k	L1	m	d4	d5	Circlip
12	26	16	19	4	16	1.1	15.2	25	16
	36				26				
	46				36				
	56				46				
	66				56				
14	26	20	25	6	16	1.3	19	30	20
	36				26				
	46				36				
	56				46				
	66				56				
	76				66				
16	26	24	28	6	16	1.3	22.9	34	24
	36				26				
	46				36				
	56				46				
	66				56				
	76				66				
	86				76				
20	26	28	32	6	16	1.6	26.6	38	28
	36				26				
	46				36				
	56				46				
	66				56				
	76				66				
	86				76				
	96				86				
25	26	34	38	8	14	1.6	32.3	45	34
	36				24				
	46				34				
	56				44				
	66				54				
	76				64				
	86				74				
	96				84				
	106				94				
30	36	39	43	10	21	1.8	37	51	39
	46				31				
	56				41				
	66				51				
	76				61				
	86				71				
	96				81				
	106				91				
	116				101				
40	46	54	58	10	28	2.1	51	67	54
	56				38				
	66				48				
	76				58				
	86				68				
	96				78				
	106				88				
	116				98				
	126				108				
	136				118				
	146				128				

The **G77** is delivered with the circlip included - no need to order the circlip separately.



External Circlip

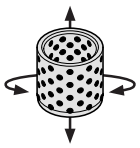
Code: **DIN471**



d1	S	a	b	d2	d3	d4	m
16	1	3.5	2.2	15.2	14.7	24.4	1.1
20	1.2	4	2.6	19	18.5	29	1.3
21		4.1	2.7	20	19.5	30.2	
22		4.2	2.8	21	20.5	31.4	
24		4.4	3	22.9	22.2	33.8	
25		4.4	3	23.9	23.2	34.8	
26		4.5	3.1	24.9	24.2	36.1	
28	1.5	4.7	3.2	26.6	25.9	38.4	1.6
29		4.8	3.4	27.6	26.9	39.6	
30		5	3.5	28.6	27.9	41	
34		5.4	3.8	32.3	31.5	45.8	
37	1.75	5.7	4.1	35	34.2	49.4	1.85
39		5.8	4.2	37	35.2	50.6	
45		6.7	4.7	42.5	41.5	59.4	
50	2	6.9	5.1	47	45.8	64.8	2.15
54		7.1	5.3	51	49.8	70	

Order: **DIN471.** d1 x S





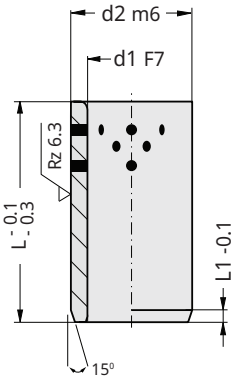
Working Direction

Guide Bush - Plain Self-Lubricating

Code: **G48**

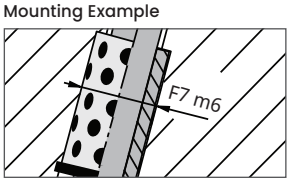
* Hole tolerance for shrink fit: H7

d1	d2	L	L1
8	12	8 10 15	2
10	14	10 12 15 20	2
12	18	10 12 15 20 25 30	2
15 16	22	12 16 20 25 30 35 40	2
18 20	24 28 30	20 25 30 35 40 50	4
24 25	33 35	20 25 30 35 40 50	4
35 30	38 40 45	20 25 30 35 40 50 60	4
38 40	50 55	25 30 35 40 50 60 80	4
42 45	55 60	30 35 40 50 60	4



d1	d2	L	L1
48 50	60 62 65	30 40 50 60 70 80 100	4
60	74 75	40 50 60 80 100	4
80	96 100	40 60 80 100 120 140	4
100 120 150 160	120 140 170 180	50 60 80 100 120 140 160	4

Mounting Example



Order: **G48.** d1 x d2 x L

Material: Bronze + graphite inserts

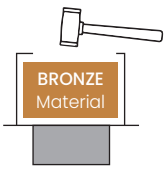




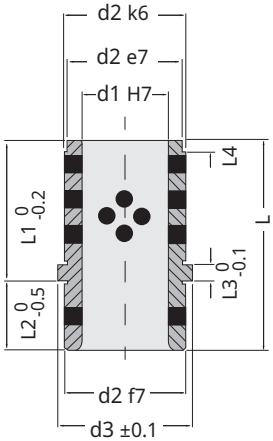
Working Direction

Guide Bush with Centre Collar Self-Lubricating

Code: **G26**



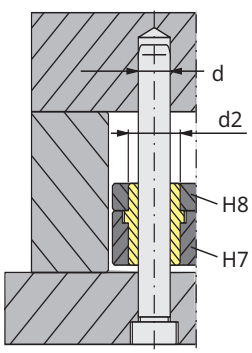
Positioning :
Pls. do not make direct fitting. Precision assembly recommended with bronze / brass.

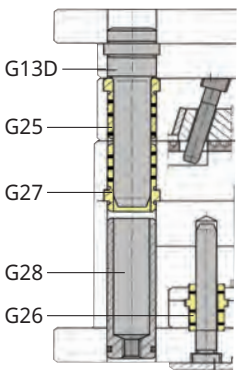


d1	L	d2	d3	L1	L2	L3	L4
NEW 10	21	14	17	12	9	6	2
NEW 12	26	16	19	17	9	6	2
14	26	20	25	17	9	6	2
16	26	20	25	17	9	6	2
20	39	26	31	22	17	6	2
22	49	30	35	27	22	6	4
25	49	30	35	27	22	6	2
30	63	42	47	36	27	6	2
40	63	50	60	36	27	8	6
50	92	63	72	55	37	8	6

Especially, it can be used in pillar systems of the ejector plates in injection moulds with Guide Pillar without oil groove **(G17)** Guide bushes are self-lubricating systems.
Heat Resistance: 150° C. It can even work in humid environments.

Mounting Example





Order: **G26.** d1 x L

Material: Bronze + graphite inserts

"Special Series for Injection Moulds"

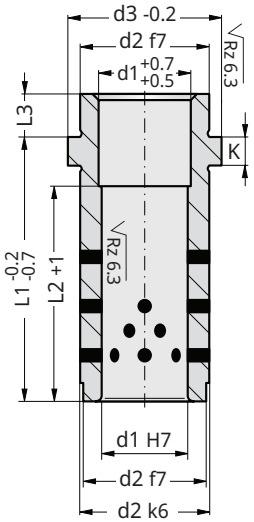




Working Direction

Guide Bush with Collar Self-Lubricating

Code: **G27**



G27 products used in injection moulds with **G13D** guide pillars (without oil groove). Higher stability is achieved with centering of the mould in length. This way, a long-term and precision working environment with other equipment (ejector - core etc.) **H7** tolerance is recommended in guide pillar / bush systems. It is compatible with the sudden movements during working. These products are not affected by vibration / impacts.

Heat resistance: 150^0 c.

d1	L1	d2	d3	K	L3
14	17	20	25	6	6
	22				
	27				
15	36				
	46				
	56				
		L2: > 2 - 3 x d1			
16	17	20	25	6	8
	22				
	27				
	36				
	46				
	56				
	66				
	76				
		L2: > 2 - 3 x d1			
20	17	26	31	6	8
	22				
	27				
	36				
18	46				
	56				
	66				
	76				
	86				
	96				
		L2: > 2 - 3 x d1			

d1	L1	d2	d3	K	L3
25	22	30	35	6	8
	27				
	36				
	46				
22	56				
	66				
	76				
24	86				
	96				
	116				
		L2: > 2 - 3 x d1			
30	27	42	47	6	8
	36				
	46				
	56				
32	66				
	76				
	86				
	96				
	116				
	136				
		L2: > 2 - 3 x d1			
40	46	54	60	10	12
	56				
	66				
	76				
42	86				
	96				
	116				
	136				
	156				
	196				
		L2: > 2 - 3 x d1			

Sliding bushing systems provide higher stability.
These products are lubricated once as a thin layer and this reduces negative effects.

Order: **G27.** d1 x L1

Material: Bronze + graphite inserts

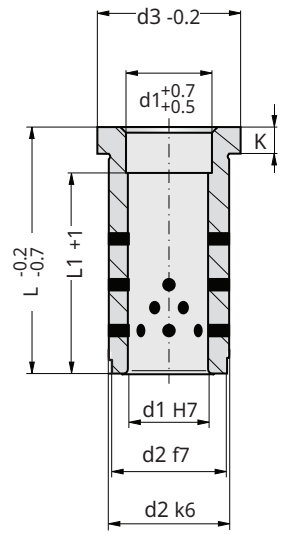




Working Direction

Guide Bush Headed Self-Lubricating

Code: **G25**



G25 products used in injection moulds with **G12D and G50D** guide pillars (without oil groove).
H7 tolerance is recommended in guide bush systems. It is compatible with the sudden movements during working. These products are not affected by vibration / Impacts.

Heat resistance: 150^0 c.
It can even work in humid environments.

d1	L	d2	d3	K
12	16	18	22	4
	21			
	27			
	36			
		L1: 2 - 3 x d1		
16	27	22	27	6
15	36			
	46			
	56			
		L1: 2 - 3 x d1		
20	27	26	31	6
	36			
	46			
18	56			
	66			
	76			
	96			
		L1: 2 - 3 x d1		
25	36	30	35	6
	46			
	56			
22	66			
	76			
24	86			
	96			
		L1: 2 - 3 x d1		

d1	L1	d2	d3	K
30	36	42	47	6
	46			
	56			
	66			
32	76			
	86			
	96			
	116			
		L1: 2 - 3 x d1		
40	36	54	60	10
	46			
	56			
	66			
	76			
42	86			
	96			
	116			
	136			
		L1: 2 - 3 x d1		
50	76	66	72	10
	86			
	96			
	116			
	136			
	156			
	196			
		L1: 2 - 3 x d1		
60	96	80	86	20
	116			
	136			
	156			
	196			
		L1: 2 - 3 x d1		

Order: **G25.** d1 x L

Material: Bronze + graphite inserts

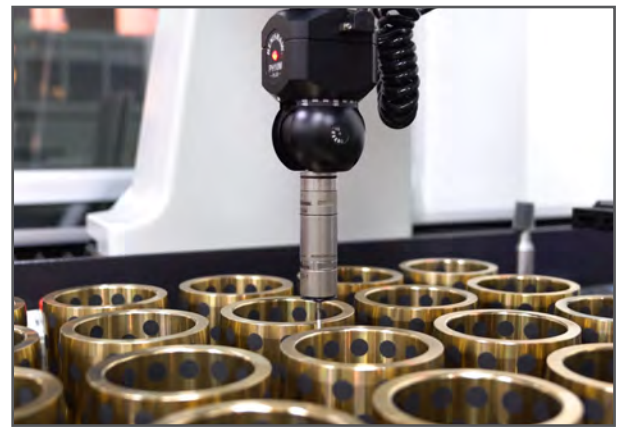
Stable and reliable Die & Mould production is obtained with components having a reliable quality. TECINMAK tends towards the highest quality standards with all development and test efforts.



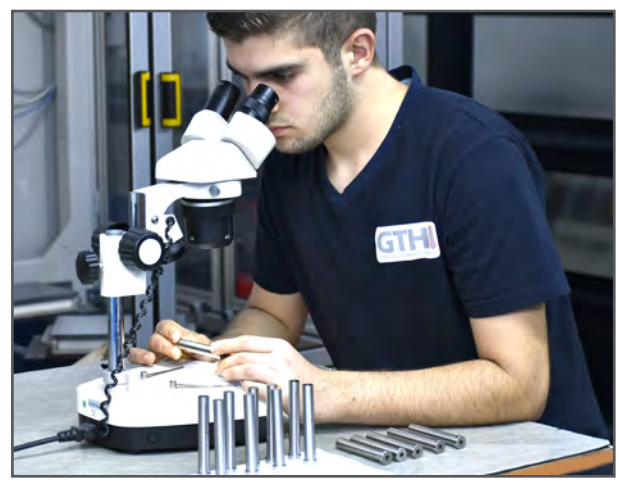






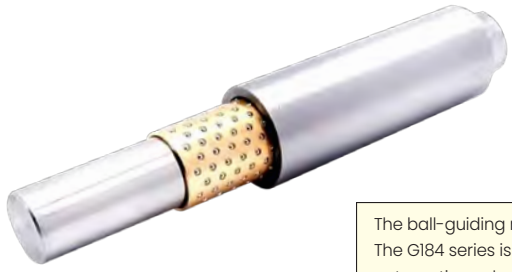






TECINMAK Reliability and Excellence in the Tooling Industry TECINMAK is a well-known manufacturer in the industry, recognized for its wide product range and fast distribution systems. TECINMAK's meticulous Quality and Control Processes ensure the highest quality at every stage of the production process. Strict controls are applied at each step, from raw material selection to the final product formation. To ensure precision shaping quality, parameters such as circularity, concentricity, flatness, and perpendicularity are carefully monitored through 3D inspection technology. Additionally, hardness tests, along with hardened and induction parameter records, are conducted at our facility. Thanks to these processes, we continue to offer a wide variety of high-quality, excellent products with the lowest tolerances under the TECINMAK Quality Assurance Labels.

NEW



Code: **G184-1**

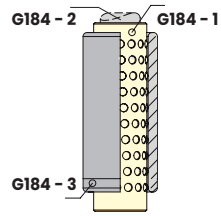
MINI Ball Cage

Code: **G184-2**

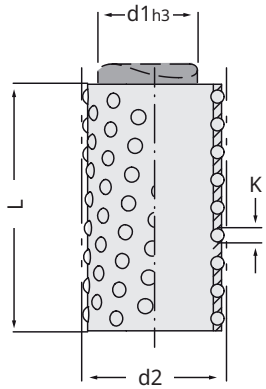
MINI Guide Pillar

Code: **G184-3**

MINI Guide Bush for G184-1



The ball-guiding mini-series enables achieving maximum precision with smooth sliding. The G184 series is widely used in high-speed stamping, injection moulding, control instruments, automation, robotics, and the semiconductor industries. Additionally, we can provide ball cages, bushes, and pillars made to custom specifications based on drawings.



Code: **G184-1**

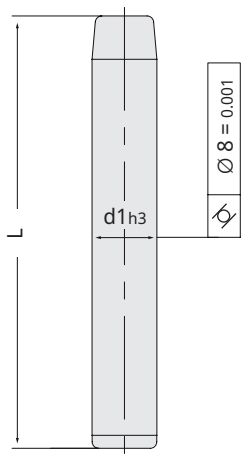
MINI Ball Cage

Material: Bronze Cage B14
Balls: Steel for bearings 100Cr6
Superfinished grade AAA

Order:
G184-1. d1 x L

d1	L	d2	K	No. of Balls
4	10	6	1	28
	15			42
	20			63
	25			77
5	10	7	1	32
	15			48
	20			72
	25			88
	30			112

d1	L	d2	K	No. of Balls
6	10	8	1	36
	15			54
	20			81
	25			99
	30			126
8	15	10	1	60
	20			90
	25			110
	30			140
	40			190



Code: **G184-2**

MINI Guide Pillar
DIN 9825 / ISO 9182

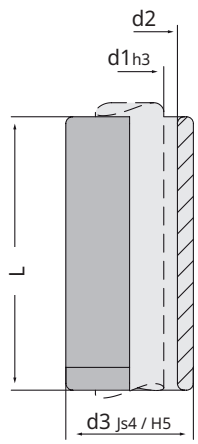
Material: 1.7131
Hardness: 58-62 HRC
Heat treatment: Case-hardening and tempering
Note: Fine grinding and super finishing

Order:
G184-2. d1 x L

d1	L
4	40
	50
	60
	80
	100
5	40
	50
	60
	80
	100

d1	L
6	40
	50
	60
	80
	100
	125
	140
	160

d1	L
8	50
	60
	80
	100
	125
	140
	160



Code: **G184-3**

MINI Guide Bush for Ball Cage

Material: 1.7131
Hardness: 60-64 HRC
Heat treatment: Case-hardening and tempering
Note: Fine grinding and super finishing

Order:
G184-3. d1 x L

d1	d2	d3	L
4	6	8	10
			15
			20
			25
5	7	10	10
			15
			20
			25
			30

d1	d2	d3	L
6	8	11	15
			20
			25
			30
			35
8	10	14	15
			20
			25
			30
			35
			40

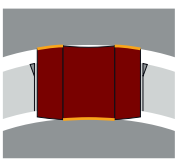
NEW



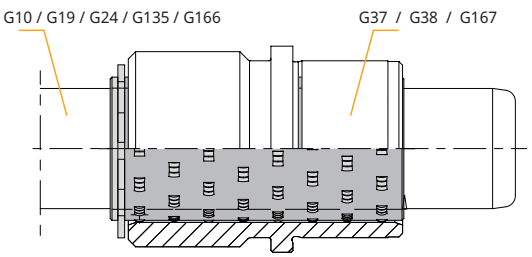
Roller Cage

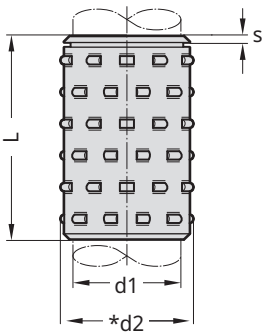
Code: **G81**

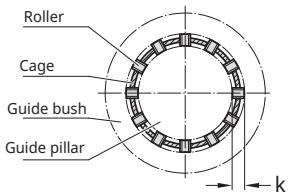
G81 Roller Cages create a linear contact with the guide bush and guide pillar, resulting in a significantly higher load-carrying capacity for each roller compared to a ball-cage of the same diameter. The profile rollers are arranged in a spiral pattern along the axis, ensuring each roller moves along its own unique path. The cages are designed with grooves to accommodate a DIN 471 circlip.



- Ideal for high-speed operations
- Suitable for applications requiring maximum precision
- Offers greater stiffness compared to ball cages
- Well-suited for **Fine Blanking & Progressive Dies** due to its ability to withstand high shear forces
- Suitable for **IML Injection Moulds**
- Reduced wear due to the axial offset of the rollers
- Optimized inlet geometry ensures smooth roller entry
- Play-free guidance achieved through optimal preload
- Can be used for axial motions only







*d2 = d1 + x2 k

d1	L	No. of Rollers	k	s
20	45	32	3	1.6
	55	40		
	65	48		
	75	56		
	85	64		
	95	72		
	105	80		
25	45	40	3	1.6
	55	50		
	65	60		
	75	70		
	85	80		
	95	90		
	105	100		
	115	110		
	125	120		
30	45	48	4	2.1
	55	60		
	65	72		
	75	84		
	85	96		
	95	108		
	105	120		
	115	132		
	125	144		
	135	156		
	145	168		
	155	180		
	165	192		

d1	L	No. of Rollers	k	s
40	55	70	4	2.1
	65	84		
	75	98		
	85	112		
	95	126		
	105	140		
	115	154		
	125	168		
	135	182		
	145	196		
	155	210		
	165	224		
	175	238		
	185	252		
	205	280		
50	65	108	4	2.1
	75	126		
	85	144		
	95	162		
	105	180		
	115	198		
	125	216		
	135	234		
	145	252		
	155	270		
	165	288		
	175	306		
	185	324		
	205	360		
63	75	154	4	2.1
	85	176		
	95	198		
	105	220		
	115	242		
	125	264		
	135	286		
	145	308		
	155	330		
	165	352		
	175	374		
	185	396		
	205	440		

Roller Cage: Aluminium
Rollers: Steel hardened, 100 Cr6, DIN 5402

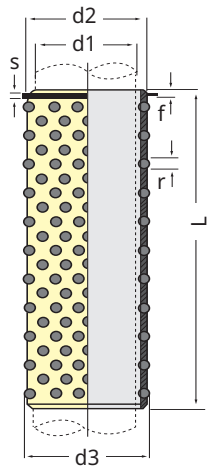
Order: **G81.** d1 x L



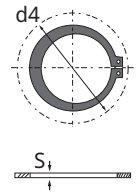
Ball Cage

Code: **G70**

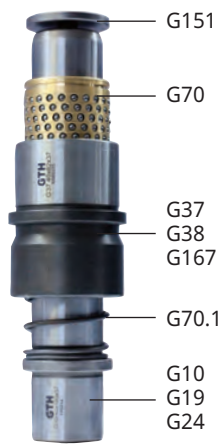
Heat resistance of ball cage is 120º C.
Ball cages are presented with circlips.



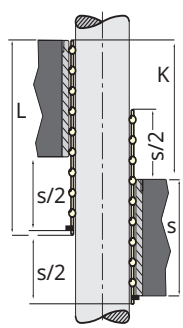




Mounting Example







L = K + 0.5 x s

Compatible Bushes





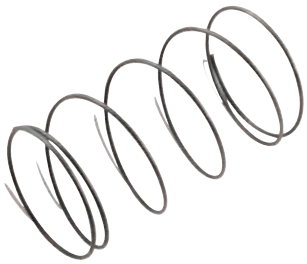


d1	L	d2	Ball		Circlip		
			d3	r	d4	f	s
NEW 10	30	13	14	2	20.6	3.2	1.2
	40						
12	40	15	16	2	22.6	3.2	1.2
	56						
15	45	20	21	3	28.4	5	2
	65						
16	71	21	22		29.6		
18	45	23	24	3	32.2	5	2
	56						
	71						
19	45	24	25	3	33.2	5	2
	56						
	71						
20	80	25	26		34.2		
	95						
24	45	29	30	3	39.1	5	2
	56						
	71						
	80	30	31		40.5		
25	95						
	105						
	120						
30	45	39	40	4	49	5	2
	56						
	75						
	80						
32	95	37	38		51.4		
	105						
	120						
	140						

d1	L	d2	Ball		Circlip		
			d3	r	d4	f	s
38	56	45	46	4	59.1	5	2
	71						
	80						
	95						
40	105	47	48		60.8		
	120						
	140						
	160						
48	71	55	56	4	70.2	5	2
	80						
	95						
	105						
50	120	57	58		72.6		
	140						
	160						
60	85	67	68	4	83.1	5.7	2.7
	95						
	105						
63	120	70	71		87		
	140						
	160						
80	105	90	92	6	108.5	6.7	3.2
	120						
	140						
	160						
	200						

Order: **G70.** d1 x L

Material: Ms58 **Ball:** 100Cr6
Tolerance: +0.001



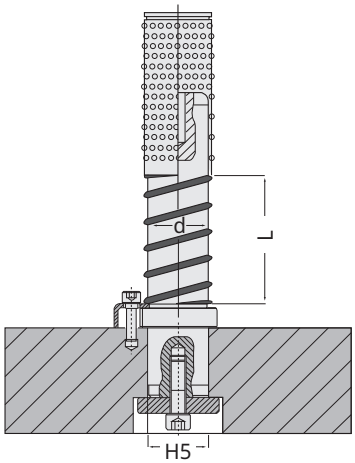
Code: **G70 - 1**

Ball Cage Retaining Spring

Note : Production in ball systems or other die inner designs Upon request.
L : 10 mm ranges

d	L
19 20	Up to 40 ~ 140
24 25	Up to 40 ~ 180
30 32	Up to 50 ~ 230
38 40	Up to 60 ~ 280
48 50	Up to 70 ~ 280
60 63	Up to 80 ~ 250

Order: **G70-1.** d x L



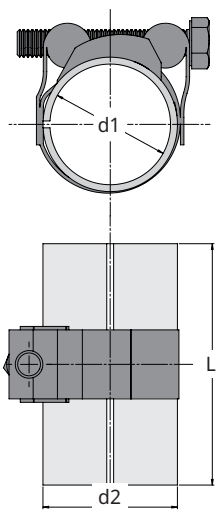
To avoid hanging at usage of ball cage bush, ball cage retaining spring can be used.



Limiting Bush

Code: **G168**

Order Code	d1	Guide Pillar	d2	L
G16816	16	15 / 16	20	20
G16820	20	19 / 20	30	20
G16825	25	24 / 25	30	30
G16832	32	30 / 32	40	30
G16840	40	38 / 40	50	40
G16850	50	48 / 50	60	40
G16863	63	60 / 63	70	40



Mounting Example



It helps to position the working length of the bushes running on the guide pillars.

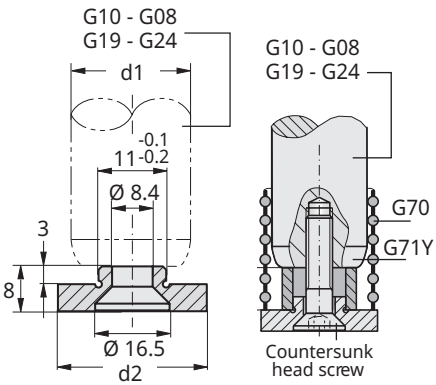




Code: **G151**

Ball Cage Holder Flange

(Locking flange for ball cage)



d1	d2	Pillar Dia.
20	25	Ø 20
25	30	Ø 25
30	37	Ø 30
40	47	Ø 40
50	57	Ø 50
60	67	Ø 60
63	71	Ø 63

Order: **G151.** d1

Material: 1.7131 (16MnCr5)

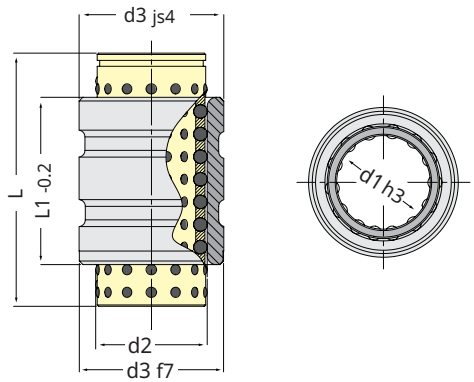
Holder Flange / (locking flange for ball cage) In ball cage bush systems, it prevents detaching of cage **(G70)** from guide pillar during operating. It is practical and provides convenience in die mounting. **In mounting to die:** Spacer tube **(G71Y)** of ball cage stroke distance and countersunk head screw **(HIC)** can be adjusted in desired length.



Plain Guide Bush for Ball Cage

Code: **G167**

ISO 9448-3



d1	d2	d3	L1	L
NEW 10	14	22	23	30
			30	40
			37	40
12	16	22	23	40
			30	40
			37	40
15 16	21 22	28	23	45
			30	45
			37	45
			47	56
			60	71

d1	d2	d3	L1	L
19 20	25 26	32	23	45
			30	45
			37	45
			47	56
			60	71
			77	95
24 25	30 31	40	23	45
			30	45
			37	45
			47	56
			60	71
			77	95
30 32	38 40	48	30	45
			37	50
			47	56
			60	71
			77	95
			95	120
38 40	46 48	58	30	45
			37	50
			47	63
			60	80
			77	95
			95	120
			120	140
48 50	56 58	70	37	50
			47	63
			60	80
			77	95
			95	120
			120	140
60 63	68 71	85	60	95
			77	95
			95	120
			120	140
80	92	105	120	140

Note: Ball Cage "G70" must be ordered separately.

Order: **G167.** d1 x L1

Material: 1.7131
Hardness: 60 - 64 HRC

The inner diameter of the products is honed, the outer diameter is precision ground. The grooves on the product are made to provide better adhesion to the hole during mounting. **Advantages of G167:** No problem finding the position when changing the bushes. High accuracy and robustness. **Mounting Instruction:** Please secure with adhesive *(Order Code: GL.64832)* Do not press-fit.



Adhesive / Loctite 648

The adhesive provides optimum push retention for **"G167"** bushes. For adhesion of bearings bushes, screws and other machine components. Very fast adhesion to handling strength.

Heat Resistance: -55⁰ / +175⁰ C **Package:** 50ml / 55 g.

Order: **GL.64832**



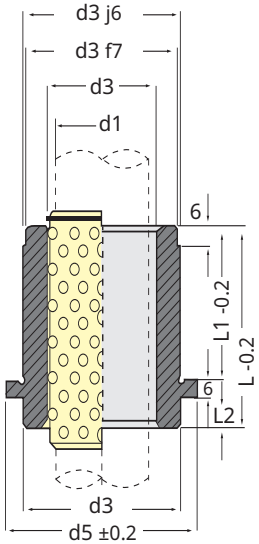




Guide Bush for Ball Cage

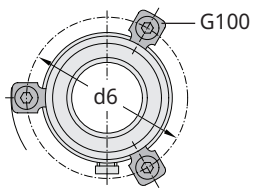
Code: **G38**

(Short type)



G38 guide bushes work without problem at dies, cuttings under 0.5 mm sheet metal thickness and where cutting space is low and when 200 stamp per minute are exceeded. It is not resistant to the lateral loads. They are preferred in cases that precison should be increased.

4 pcs. clamp for **d1: 38** mm and larger sizes.



d1	L	d2	d3	d5	d6	L1	L2
15 16	35	21 22	30	34	46.2	23	12
19 20	35	25 26	32	40	52.4	23	12
24 25	35	30 31	40	48	60.8	23	12
30 32	42	37 40	48	56	67.8	30	12
38 40	52	46 48	58	66	78	37	15
48 50	65	56 58	70	80	93.2	47	18
60 63	80	68 71	85	95	108.4	60	20
80	80	92	105	118	131.5	60	20

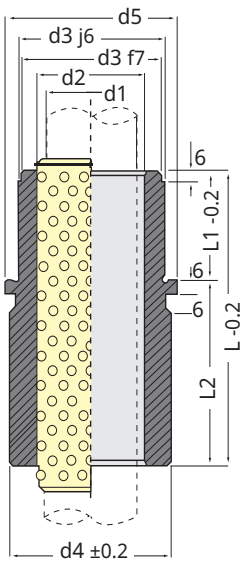
Order:
G38. d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

Guide Bush for Ball Cage

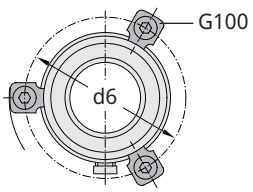
Code: **G37**

(Long type)



High precision guide bushes for ball cage: Despite providing high reliability at high speeds, contacting balls from one point, brings the system to precision status against impacts and side loads. This disadvantage is compensated to give product diameters up to a certain point in the upper tolerance, the working surface should be precise.

4 pcs. clamp for **d1: 38** mm and larger sizes.



d1	L	d2	d3	d4	d5	d6	L1	L2
15 16	43 59	21 22	30	32	34	46.2	23	20 36
19 20	43 59	25 26	32	39	40	52.4	23	20 36
24 25	59 79	30 31	40	46	48	60.8	23	36 56
30 32	75 93	38 40	48	53	56	67.8	30	45 63
38 40	82 108	46 48	58	63	66	78	37	45 71
48 50	97 127	56 58	70	77	80	93.2	47	50 80
60 63	116 150	68 71	85	92	95	108.4	60	56 90
80	120 150	92	105	115	118	131.5	60	60 90

Order:
G37. d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

Note: Two different products in section (d1) are to avoid reverse closing of dies during mounting by using three pcs. main dimensions and one pcs. auxiliary **(d1= Ø15 -19 - 24 - 32 - 38 - 48 - 60 mm)** dimensions at same die, is avoided reverse closing of die plate. Thus, symmetrical die closure is ensured. **Example:** In the same die (Ø20 x 3 - Ø19 x 1) = 4 pieces **Note:** Should be ordered with G70.

NEW







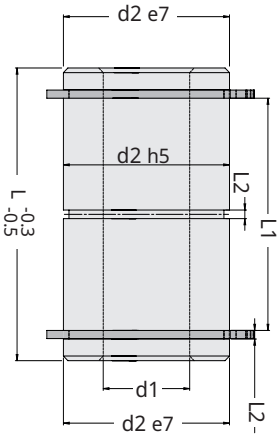
interior view

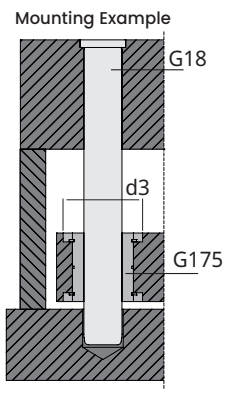
NEW



Linear Ball Bearing Bush with Circlips

Code: **G175**



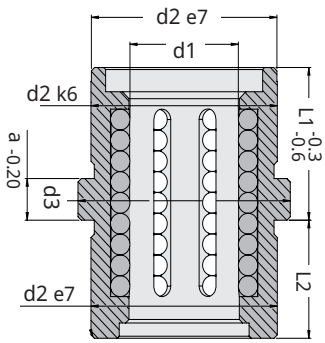


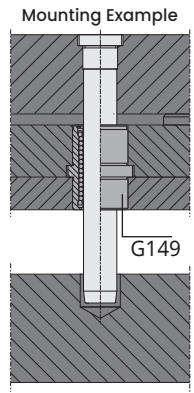
Operating Temp.: Max. 200°

Order Code	d1	L	d2	L1	L2	d3	Ball Dia.	Line Pcs.
G1751226	12	26	24	17	1.2	34	3	5
G1751834	18	34	30	25	1.5	41	3	6
G1751844	18	44	30	35	1.5	41	3	6
G1752034	20	34	32	25	1.5	43	3	6
G1752044	20	44	32	35	1.5	43	3	6
G1752544	25	44	40	30	1.75	51	3	8
G1752554	25	54	40	40	1.75	51	3	8
G1753054	30	54	48	40	1.75	61	4	8
G1753063	30	63	48	50	1.75	61	4	8

Linear Ball Bearing Bush with Centre Collar

Code: **G149**





Operating Temp.: Max. 200°

d1	L1	L2	d2	d3	a	Ball Dia.	Line Pcs.
10	12 17	9	20	24	6	2.5	5
12	17	9 12	24	28	6	3	5
18	22	12 17	30	34	6	3	6
20	22	17	32	36	6	3	6
25	27	17 22	40	45	6	3	8
30	32 36	22 27	48	54	8	4	8
40	36	27	60	66	8	4	8

Order:
G149. d1 x L1 x L2

3D CAD FILES GUVENAL

TECINMAK linear Ball Bushings are designed for high-performance linear motion in demanding tooling and mould making applications. Each bushing features a precision-ground steel outer shell and a durable bronze internal cage that guides multiple rows of recirculating ball bearings. Unlike conventional designs, the rolling elements in TECINMAK bushes move at a slight angle relative to the axis of motion. This angled path increases the contact area with the pillar, significantly enhancing load capacity and operational stability. The all-metal construction ensures high resistance to heat, wear, and contamination—making these bushings ideal for environments where thermal and mechanical stress are common. TECINMAK bushes are available in four different outer geometries, all sharing the same internal ball guidance system to ensure consistent performance across configurations.





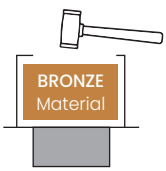
interior view



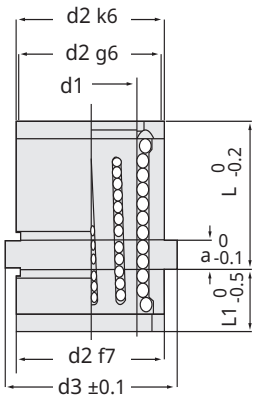
Linear Ball Bearing Bush

Code: **G75**

Free stroke, rapid and limitless movement.
New for ejector plate.
Operating Temp.: Max. 200⁰



Positioning :
Pls. do not make direct fitting.
Precision assembly recommended with bronze / brass.



d1	L	d2	d3	a	L1	Ball Dia.	Line Pcs.
12	22	24	28	6	8	3	5
18	26	30	34	6	9	3	6
20	26	32	36	6	9	3	6
	36						
25	26	40	45	6	10	3	8
	36						
	46						
30	46	48	54	8	12	4	8
	66						
40	46	60	66	8	12	4	8
	66						
NEW 50	56	70	76	10	25	4	10

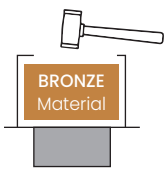
Order: **G75.** d1 x L

3D CAD FILES GUVENAL

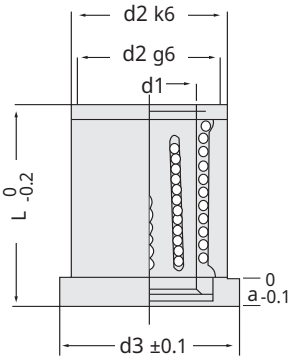
Linear Ball Bearing Bush Headed

Code: **G74**

Balls are worked independently by rotating, they are not frictional.
Operating Temp.: Max. 200⁰



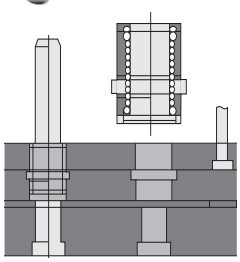
Positioning :
Pls. do not make direct fitting.
Precision assembly recommended with bronze / brass.



d1	L	d2	d3	a	Ball Dia.	Line Pcs.
18	35	30	34	6	3	6
20	35	32	36	6	3	6
	45					
	55					
30	45	48	54	8	4	8
	63					
40	45	60	66	8	4	8
	63					



G18 Compatible Pillar



Order: **G74.** d1 x L

3D CAD FILES GUVENAL

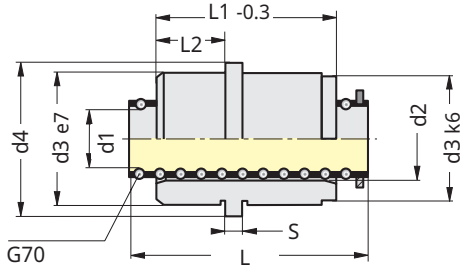
New Linear Ball Bearing Bushes G74 - G75 - G149 - G175: There are balls at inclined - serial - parallel intermediate channels - lines inside of linear **G74-G75-G149-G175** bushes (balls move by rotating in slot and frictionless). In this way, axial motion remains limitlessly free. Also, bearing logic is provided to the steel bush. **The advantage of our new product G75 according to other similar product G29:** Extra stroke distances at unlimited axial direction and robust powerful centering is provided - free motion unlimited with low tolerance absorbs other pinking and tauting. It is presented unlimited motion advantages with compatibility to high speeds.
Linear bush G74: Quality control equipment - practical and reliable product for machine and mould production and device design.
For maintenance and cleaning: "TECINMAR Signum ***(order code: W160110)***" spray oil is used.

WARNING ! During production stage of **G74 - G75 - G149 - G175** products ball bearings are positioned at a certain angle. Other imitation products has no angle and the balls are at a single centre.



Steel Bush + Ball Cage Bush Set

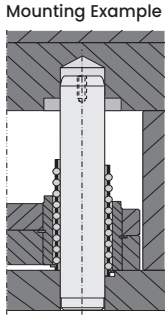
Code: **G29**



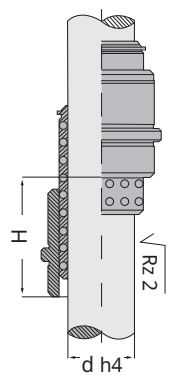
d1	L	L1	L2	d2	d3	d4	S	H
12	40	24	6	16	22	26	6	50
	56							82
18	45	34	11	24	30	35	6	44
	56							66
	71							96
30	56	54	21	38	46	52	6	32
	75							70
	95							110

It provides more precision and rapid motion at ejector plates in injection moulds.

Mounting Example



more precision and rapid motion



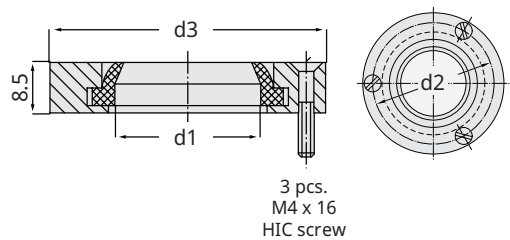
Order: **G29.** d1 x L

Ball Material: 1.3505 (100Cr6)
Cage: Ms58 **Steel Bush:** 1.7131 (16MnCr5)



Perforated Flange

Code: **G22**



Order Code	d1	d2	d3
G22 24	24	45	55
G22 25	25	45	55
G22 32	32	55	65
G22 30	30	55	65
G22 38	38	65	75
G22 40	40	65	75
G22 48	48	78	94
G22 50	50	78	94
G22 60	60	92	110
G22 63	63	92	110

Material: 1.7131 (16MnCr5)

Guide Pillar Protection Flange:

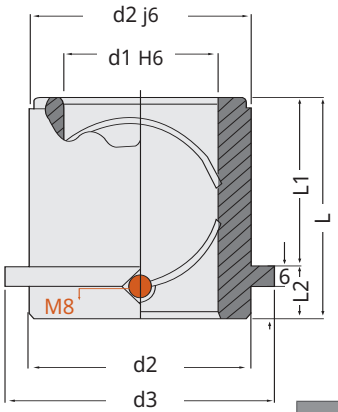
Guide pillar protection flange protects dirt to be occurred at dies or thrust etc. Different various usage areas can be created in dies.

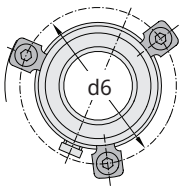


Guide Bush, Thick & Short Type

Code: **G34K**

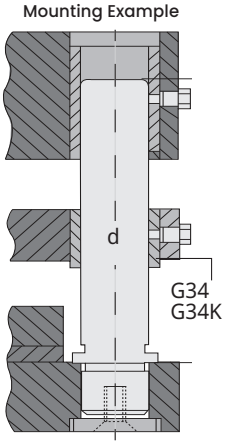
(with collar & lubricating)





4 pcs. clamp for d1 = 38 mm and larger sizes.

Mounting Example



d1	L	d2	d3	L1	L2
19 20	35	32	40	23	12
24 25	35	40	48	23	12
32 30	42	48	56	30	12
38 40	52	58	66	37	15
48 50	65	70	80	47	18
60 63	80	85	95	60	20
80	80	105	118	60	20

Order:
G34K. d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

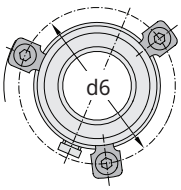
Thick type steel bushes: To provide perpendicularity at bush and guide pillar with new design of TECINMAR **Die Components** on product suitable flange fixation (balanced layout) on bearing area is created. During mounting, shrink fit should not be done.
To avoid clamping and surface disorders at products by inserting soft tight products to the slots, mounting can be ensured with clamps.



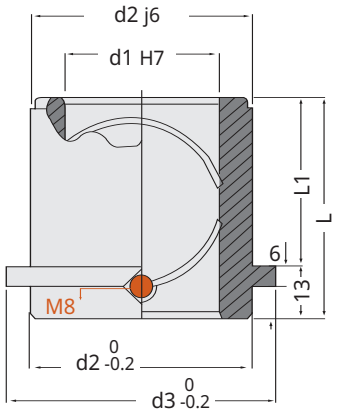
Guide Bush, Thin & Short Type

Code: **G34**

(with collar & lubricating)



4 pcs. clamp for d1 = 38 mm and larger sizes.



Mounting Details:

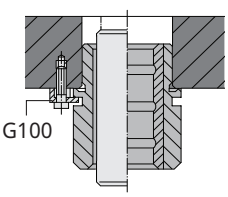
Code: **G34K**

d1	d6
20	52.4
25	60.8
30	69
40	79
50	93.2
63	108.4
80	131.5

Code: **G34**

d1	d6
16	46.2
20	46.2
25	51.4
30	56.4
40	66.8
50	77

d1	L	d2	d3	L1
16	36	28	34	23
20	36	28	34	23
25	36	34	39	23
30	50	39	44	37
40	50	50	54	37
50	60	60	64	47



Order:
G34. d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

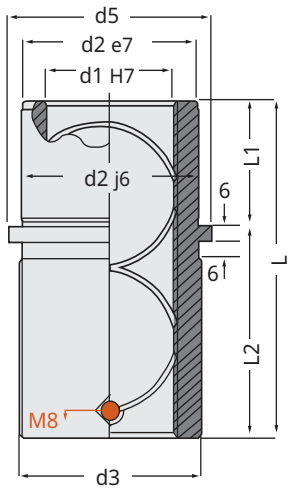
For being TECINMAR **Die Components** products having long life and maximum resistance against abrasion, unique precision has provided at surface roughness. This is also created specific bushing system.



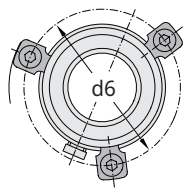
Guide Bush, Thin & Long Type

Code: **G33**

(with collar & lubricating)



G100
Clamping Detail



4 pcs. clamp for d1 = 38 mm and larger sizes.

d1	L	d2	d3	L1	L2
16	70	28	34	23	47
20	70	28	34	23	47
25	80	34	39	23	57
30	90	39	44	37	53
40	100	50	54	37	63
50	120	60	64	47	73

Mounting Details:

Code: **G33**

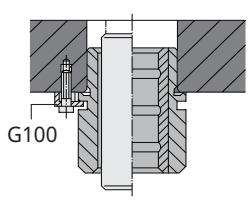
d1	d6
16	46.2
20	46.2
25	51.4
30	56.4
40	66.8
50	77

Code: **G33K**

d1	d6
20	52.4
25	60.8
30	69
40	79
50	93.2
63	108.4
80	131.5

Order:
G33 d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC



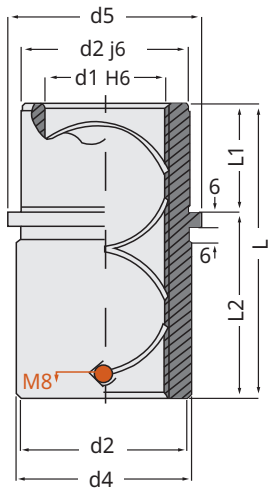
For being TECINMAR **Die Components** products having long life and maximum resistance against abrasion, unique precision has provided at surface roughness. This is also created specific bushing system.



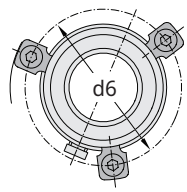
Guide Bush, Thick & Long Type

Code: **G33K**

(for heavy dies / with collar & lubricating)

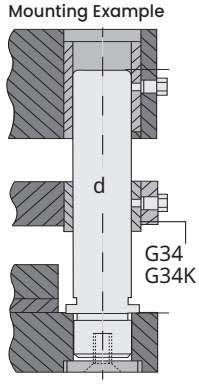


G100
Clamping Detail



4 pcs. clamp for d1 = 38 mm and larger sizes.

Mounting Example



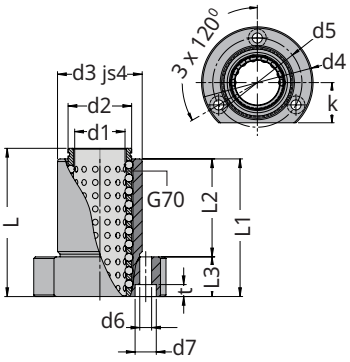
Order:
G33K d1 x L x L1

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

d1	L	d2	d4	d5	L1	L2
19	43	32	39	40	23	20
20	59					36
24	59	40	46	48	23	36
25	79					56
32	75	48	53	56	30	45
30	93					63
38	82	58	63	66	37	45
40	108					71
48	97	70	77	80	47	50
50	127					80
60	116	85	92	95	60	56
63	150					90
80	120	105	115	118	60	60
	150					90

Thick type steel bushes: Please don't use two products without oil groove together. One of product in guide pillar or bushes should be used as set with oil groove. While using guide pillar and bush sets, should be considered that both of them have same brand. Because, grinding tolerance can be different in different guide pillars and bushes. Also, this can be caused big problems during both mounting and operating. The dimensions not giving their tolerance are subjected to general tolerance (DIN 7168).





Code: **G72 – 1**

Guide Bush with Flange, for Ball Cage

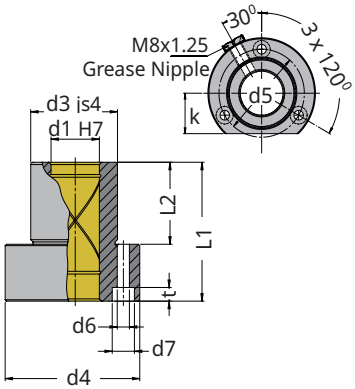
* 3 pcs **"IMB"** coded screws are used to fixed product.
* Clamping Screws & "G70" coded Ball Cage Bush should be ordered separately.

Order: **G72 – 1.** d1 x L1 x L2

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

d1	d2	d3	d4	d5	d6	d7	t	k	L	L1	L2	L3
19	25	32	50	40	4.5	8	5	18	45	38	23	15
20	26								56	45	30	
									72	52	37	
24	30	40	63	50	5.5	10	6	23	45	38	23	15
25	31								71	55	30	25
									72	62	37	
30	38	48	72	58	5.5	10	6	28	56	45	30	15
32	40								71	62	37	25
									80	72	47	
38	46	58	85	70	7	11	7	33	63	55	30	25
40	48								80	67	37	30
									95	77	47	
48	56	70	104	86	9	15	9	38	80	62	37	25
50	58								95	89	47	42
									120	102	60	
60	68	85	120	100	9	15	9	46	95	89	47	42
63	71								120	102	60	
80	92	105	148	125	12	18	12	56	140	125	75	50





Code: **G72 – 2**

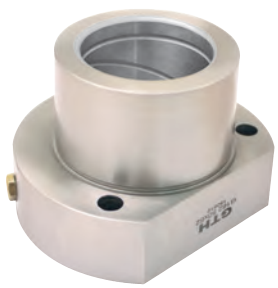
Guide Bush with Flange, Bronze Coated

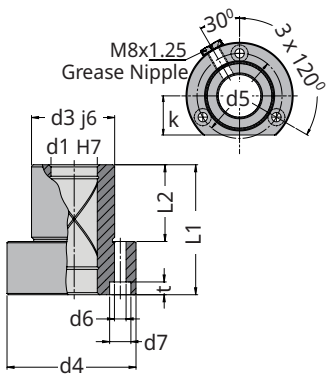
* 3 pcs **"IMB"** coded screws are used to fixed product.
* Clamping Screws should be ordered separately.

Order: **G72 – 2.** d1 x L1 x L2

Material: 1.1213 (Cf53)
Bronze Casting

d1	d3	d4	d5	d6	d7	t	k	L1	L2
19	32	50	40	4.5	8	5	18	38	23
20								45	30
								52	37
24	40	63	50	5.5	10	6	23	38	23
25								55	30
								62	37
30	48	72	58	5.5	10	6	28	45	30
32								62	37
								72	47
38	58	85	70	6.6	11	7	33	55	30
40								67	37
								77	47
48	70	104	86	9	15	9	38	62	37
50								89	47
								102	60
60	85	120	100	9	15	9	46	89	47
63								102	60
80	105	146	125	11	18	12	56	125	75





Code: **G72 – 3**

Guide Bush with Flange, Steel

* 3 pcs **"IMB"** coded screws are used to fixed product.
* Clamping Screws should be ordered separately.

Order: **G72 – 3.** d1 x L1 x L2

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

d1	d3	d4	d5	d6	d7	t	k	L1	L2
19	32	50	40	4.5	8	5	18	38	23
20								45	30
								52	37
24	40	63	50	5.5	10	6	23	38	23
25								55	30
								62	37
30	48	72	58	5.5	10	6	28	45	30
32								62	37
								72	47
38	58	85	70	6.6	11	7	33	55	30
40								67	37
								77	47
48	70	104	86	9	15	9	38	62	37
50								89	47
								102	60
60	85	120	100	9	15	9	46	89	47
63								102	60
80	105	146	125	11	18	12	56	125	75